Modelling a 3D Printer

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0 Introduction

3D printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Unlike Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a sturdier casing and faster printing speeds. Our task is to model and simulate the system.

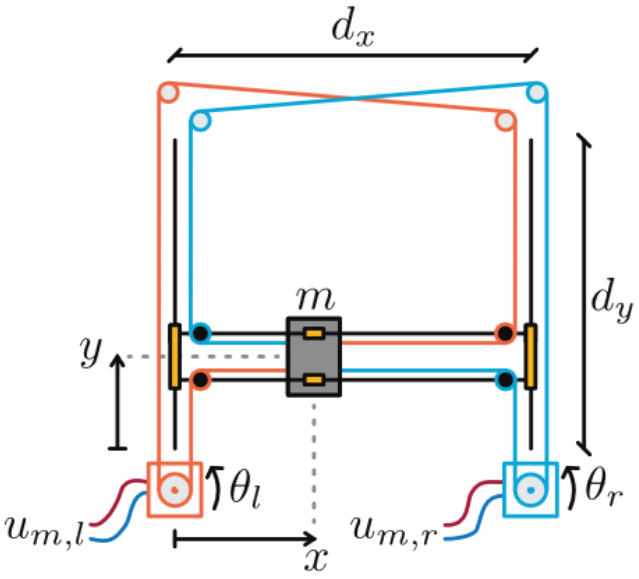


Figure 1: Schematic of the Core XY mechanism.

0.1 Symbol Table

Table 1: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_L	–
Motor voltage left	$u_{m,L}$	–
Motor current left	i_L	–
Motor radius	r_m	11 mm
Motor inductance	L_m	10 mH
Motor resistance	R_m	90 Ω
Motor inertia	J_m	8×10^{-6} kg m^2
Motor constant	K_m	0.07 N m/A
Tool head position	x, y	–
Tool head mass	m	60 g
Nozzle voltage	u_n	–
Nozzle inductor current	$i_{L,n}$	–
Nozzle capacitance	C_n	–
Nozzle inductance	L_n	–
Capacitor voltage	u_c	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	9 K/W
Nozzle resistance	R_n	8 Ω
Resistor thermal capacity	C_r	–
Resistor temperature	T_r	–
Thermal resistance to environment	R_{env}	–
Reference resistance	R_0	–
Reference temperature	T_0	–
Temperature coefficient of resistance	α	–
Nozzle temperature	T_n	240 °C
Environmental temperature	T_e	26.85 °C
Bearing friction rolling	μ_r	2.7×10^{-8} N m s/rad
Bearing friction linear x-direction	$\mu_{l,x}$	0.002 N s/m
Bearing friction linear y-direction	$\mu_{l,y}$	0.001 N s/m
Bearing radius	r_b	3 mm
Print bed temperature	T_b	105 °C
Print area dimension x	d_x	29 cm
Print area dimension y	d_y	29 cm

1 Physical Modelling

1.1 Kinematics

The first task is to derive the direct kinematics of the Core XY mechanism. This is slightly complicated by the fact that rotating one motor will move the tool head in both x and y directions. By removing the right motor, we can see that by moving the tool head in the positive x direction **or** in the positive y direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equations:

$$\begin{aligned} \Delta L &= \Delta x + \Delta y \\ \Delta R &= \Delta x - \Delta y \\ \Delta x &= \frac{\Delta L + \Delta R}{2} \\ \Delta y &= \frac{\Delta L - \Delta R}{2} \end{aligned} \tag{1.1}$$

With ΔL and ΔR being the length of belt moved by the motors, and Δx and Δy being the change in position of the tool head.

$$\begin{aligned} \Delta L &= r_m \, \theta_L \\ \Delta R &= r_m \, \theta_R \end{aligned} \tag{1.2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the tool head is at $(0, 0)$, we get the following equations:

$$\begin{aligned} x(\theta_L, \theta_R) &= \frac{r_m \theta_L + r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L + \theta_R\right) \\ y(\theta_L, \theta_R) &= \frac{r_m \theta_L - r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L - \theta_R\right) \end{aligned} \tag{1.3}$$

$$\begin{aligned} \dot{x}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L + \dot{\theta}_R\right) \\ \dot{y}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L - \dot{\theta}_R\right) \end{aligned} \tag{1.4}$$

1.2 Dynamics

In the next part we model the dynamics of the 3D printer system by applying the Euler–Lagrange equation. Additionally, we will split the problem into a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalised coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_L \\ q_R \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix} \tag{1.5}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_L \\ i_R \end{bmatrix} \text{ (electrical part)} \tag{1.6}$$

1.2.1 Mechanical Part

Kinetic Energy

Firstly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_{\text{mech}} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) \tag{1.7}$$

As the whole system is planar, we can see that there are no potential energy terms as it will be constant.

Dissipative Energy

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system, two along each axis. We define r_b as the bearing radius, later it will be kept the same as the motor radius for simplicity. From (1.3), we can get transformations (1.8) with trivial time derivatives.

$$\theta_{b,L1} = \frac{x+y}{r_b}, \qquad \theta_{b,R1} = \frac{x-y}{r_b} \tag{1.8}$$

These are used for the two fixed bearings on each belt. Then, for each side we get rotational dissipative energy:

$$D_{L1} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L1}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$
$$D_{R1} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R1}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

For the two carriage bearings, if we assume the belt to be horizontal on the carriage like Figure 1 shows them, they are rotating only when the carriage moves along the x axis.

$$\theta_{b,L2} = \frac{x}{r_b}, \qquad \theta_{b,R2} = \frac{x}{r_b} \tag{1.9}$$

With that we get two more terms:

$$D_{L2} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L2}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \dot{x}^2$$
$$D_{R2} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R2}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \dot{x}^2$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = \frac{\mu_r}{r_b^2} \left[4\dot{x}^2 + 2\dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = 2 \cdot \tfrac{1}{2}\mu_{l,x} \, \dot{x}^2 + 2 \cdot \tfrac{1}{2}\mu_{l,y} \, \dot{y}^2 = \mu_{l,x} \, \dot{x}^2 + \mu_{l,y} \, \dot{y}^2$$

Then, combining the terms we get:

$$D_{\text{mech}} = \left[4\,\frac{\mu_r}{r_b^2} + \mu_{l,x}\right] \dot{x}^2 + \left[2\,\frac{\mu_r}{r_b^2} + \mu_{l,y}\right] \dot{y}^2 \tag{1.10}$$

1.2.2 Electrical Part

Now we model the energies of the electrical part of the system. Keep in mind that both motors are the same. Below we can see the equivalent circuit of a DC motor:

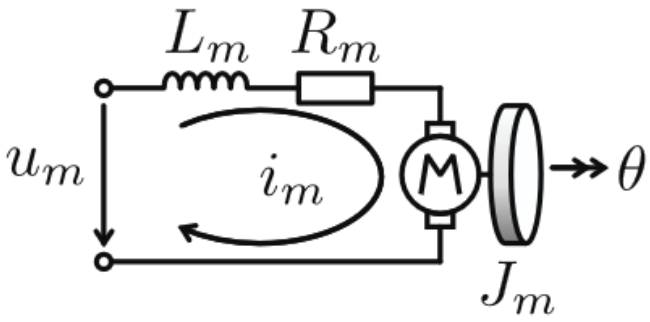


Figure 2: Equivalent circuit of a DC motor.

Kinetic Energy

The inductor and the inertial element will contribute to the kinetic energy of this part of the system. Similarly to (1.8), using the motor shaft radius:

$$K_L = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \dot{\theta}_L^2 = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$K_R = \tfrac{1}{2} i_R^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\right)$$

Combining them we get:

$$K_{\text{el}} = \tfrac{1}{2} L_m \left[i_R^2 + i_L^2\right] + J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2\right) \tag{1.11}$$

As there are no capacitors in the system, there is no potential energy change as in the mechanical part.

Dissipative Energy

The only dissipative term in the electrical part of the system is the resistor:

$$\begin{gathered} D_{\text{el},L} = \tfrac{1}{2} i_L^2 R_m, \qquad D_{\text{el},R} = \tfrac{1}{2} i_R^2 R_m \\ D_{\text{el}} = \tfrac{1}{2} R_m \left(i_L^2 + i_R^2\right) \end{gathered} \tag{1.12}$$

Matrix Equations

Finally, we realise that the system is linear, and write the matrix expressions:

$$K = \tfrac{1}{2}\, \dot{q}^\top \mathbf{M}\, \dot{q}, \qquad D = \tfrac{1}{2}\, \dot{q}^\top \mathbf{N}\, \dot{q} \tag{1.13}$$

With corresponding matrices:

$$\mathbf{M} = \begin{bmatrix} m + J_m \frac{2}{r_m^2} & 0 & 0 & 0 \\ 0 & m + J_m \frac{2}{r_m^2} & 0 & 0 \\ 0 & 0 & L_m & 0 \\ 0 & 0 & 0 & L_m \end{bmatrix}$$
$$\mathbf{N} = \begin{bmatrix} 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x} & 0 & 0 & 0 \\ 0 & 4\frac{\mu_r}{r_b^2} + 2\mu_{l,y} & 0 & 0 \\ 0 & 0 & R_m & 0 \\ 0 & 0 & 0 & R_m \end{bmatrix} \tag{1.14}$$

1.2.3 Generalised Force

Mechanical Coordinates

The non-conservative forces in the mechanical part are given by the motor torques τ_L and τ_R. The virtual work they perform is:

$$\Delta W = \tau_L \, \Delta\theta_L + \tau_R \, \Delta\theta_R$$

With the familiar transformation, analogous to (1.8) but using the motor radius r_m, we get:

$$\Delta W = \frac{\tau_L + \tau_R}{r_m} \, \Delta x + \frac{\tau_L - \tau_R}{r_m} \, \Delta y$$

Applying the motor law $\tau = K_m i$, the generalised forces for x and y are:

$$Q_x = \frac{K_m}{r_m} \left(i_L + i_R \right), \qquad Q_y = \frac{K_m}{r_m} \left(i_L - i_R \right) \tag{1.15}$$

Electrical Coordinates

The non-conservative forces in the electrical part are the input voltages and the back-EMF of the motors:

$$\Delta W = \left(u_{m,L} - K_m \dot{\theta}_L \right) \Delta q_L + \left(u_{m,R} - K_m \dot{\theta}_R \right) \Delta q_R$$

Using the same transformation, we get the generalised forces for q_L and q_R:

$$Q_{q_L} = u_{m,L} - \frac{K_m}{r_m} \left(\dot{x} + \dot{y} \right), \qquad Q_{q_R} = u_{m,R} - \frac{K_m}{r_m} \left(\dot{x} - \dot{y} \right) \tag{1.16}$$

Lagrange Equations

Using the fact that this is a linear system and there are no potential energies we get:

$$\mathcal{L} = K - V = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q} - V, \qquad V = 0 \quad \Rightarrow \quad \mathcal{L} = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{d}{dt} \left(\mathbf{M} \dot{q} \right) = \mathbf{M} \ddot{q}$$
$$\frac{\partial \mathcal{L}}{\partial q} = 0$$
$$\frac{\partial D}{\partial \dot{q}} = \mathbf{N} \dot{q}$$

And finally solving the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \mathbf{Q}, \qquad \mathbf{Q} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{q_L} \\ Q_{q_R} \end{bmatrix}$$

$$\mathbf{M}\ddot{q} + \mathbf{N}\dot{q} = \mathbf{Q} \tag{1.17}$$

Expanding back out of matrix form we get:

$$\begin{aligned}
\left(m + J_m \frac{2}{r_m^2}\right)\ddot{x} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}\right)\dot{x} - \frac{K_m}{r_m} i_L - \frac{K_m}{r_m} i_R &= 0 \\
\left(m + J_m \frac{2}{r_m^2}\right)\ddot{y} + \left(4\frac{\mu_r}{r_b^2} + 2\mu_{l,y}\right)\dot{y} - \frac{K_m}{r_m} i_L + \frac{K_m}{r_m} i_R &= 0 \\
L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\dot{x} + \frac{K_m}{r_m}\dot{y} - u_{m,L} &= 0 \\
L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\dot{x} - \frac{K_m}{r_m}\dot{y} - u_{m,R} &= 0
\end{aligned}$$

Or with substitutions, we get the final form of the four differential equations:

$$m_{\text{eq}} = m + \frac{2J_m}{r_m^2}, \qquad b_x = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}, \qquad b_y = 4\frac{\mu_r}{r_b^2} + 2\mu_{l,y}$$

$$\boxed{\begin{aligned}
\ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L + i_R\right) \\
\ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L - i_R\right) \\
\frac{di_L}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} + \dot{y}\right) - \frac{R_m}{L_m} i_L + \frac{u_{m,L}}{L_m} \\
\frac{di_R}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} - \dot{y}\right) - \frac{R_m}{L_m} i_R + \frac{u_{m,R}}{L_m}
\end{aligned}} \tag{1.18}$$

Or in **state-space** form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix}, \qquad \dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \\ \dfrac{di_L}{dt} \\ \dfrac{di_R}{dt} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_{m,L} \\ u_{m,R} \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \tag{1.19}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b_x}{m_{\text{eq}}} & 0 & \frac{K_m}{m_{\text{eq}} r_m} & \frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & -\frac{b_y}{m_{\text{eq}}} & \frac{K_m}{m_{\text{eq}} r_m} & -\frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & -\frac{K_m}{r_m L_m} & -\frac{K_m}{r_m L_m} & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & -\frac{K_m}{r_m L_m} & \frac{K_m}{r_m L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix}$$

1.2.4 Decoupling the x and y Dynamics

To decouple the system, we introduce a set of useful substitutions:

$$\boxed{\begin{aligned} i_x &= i_L + i_R, & u_{m,x} &= u_{m,L} + u_{m,R} \\ i_y &= i_L - i_R, & u_{m,y} &= u_{m,L} - u_{m,R} \\ i_L &= \frac{i_x + i_y}{2}, & u_{m,L} &= \frac{u_{m,x} + u_{m,y}}{2} \\ i_R &= \frac{i_x - i_y}{2}, & u_{m,R} &= \frac{u_{m,x} - u_{m,y}}{2} \end{aligned}} \tag{1.20}$$

By substituting our new decoupled currents we get:

$$\begin{aligned} \frac{di_L}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} + \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} - \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x - \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} + u_{m,y}\right) \\ \frac{di_R}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} - \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} + \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x + \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} - u_{m,y}\right) \end{aligned}$$

Then by adding and subtracting the two equations to cancel terms we get:

$$\begin{aligned} \frac{di_x}{dt} &= -\frac{2K_m}{r_m L_m}\dot{x} - \frac{R_m}{L_m} i_x + \frac{1}{L_m} u_{m,x} \\ \frac{di_y}{dt} &= -\frac{2K_m}{r_m L_m}\dot{y} - \frac{R_m}{L_m} i_y + \frac{1}{L_m} u_{m,y} \end{aligned} \tag{1.21}$$

And by substituting (1.20) into the first two equations of (1.18):

$$\begin{aligned} \ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m} i_x \\ \ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m} i_y \end{aligned} \tag{1.22}$$

And finally:

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ \dot{x} \\ i_x \\ y \\ \dot{y} \\ i_y \end{bmatrix}, \qquad \dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dfrac{di_x}{dt} \\ \dot{y} \\ \ddot{y} \\ \dfrac{di_y}{dt} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \begin{bmatrix} u_{m,x} \\ u_{m,y} \end{bmatrix}$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\,\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\,\tilde{\mathbf{u}} \tag{1.23}$$

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\dfrac{b_x}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} & 0 & 0 & 0 \\ 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{b_y}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} \end{array}\right], \qquad \tilde{\mathbf{B}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{L_m} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{L_m} \end{bmatrix}$$

1.3 Drawing Shapes

Rearranging the decoupled equations of motion (1.22), we can express the currents needed to follow a prescribed motion of the tool head:

$$\boxed{\begin{aligned} i_x &= \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{x} + \frac{b_x r_m}{K_m}\,\dot{x} \\ i_y &= \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{y} + \frac{b_y r_m}{K_m}\,\dot{y} \end{aligned}} \tag{1.24}$$

The motor equations for each motor are:

$$\begin{aligned} u_{m,L} &= L_m \frac{di_L}{dt} + R_m i_L + K_m \dot{\theta}_L = L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\,(\dot{x} + \dot{y}) \\ u_{m,R} &= L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\,(\dot{x} - \dot{y}) \end{aligned}$$

Adding them, and using (1.20):

$$u_{m,x} = u_{m,L} + u_{m,R} = L_m \left[\frac{di_L}{dt} + \frac{di_R}{dt}\right] + R_m\,[i_L + i_R] + \frac{2K_m}{r_m}\,\dot{x}$$

and similarly by subtracting them, we get the decoupled input voltages, consistent with (1.21):

$$\boxed{\begin{aligned} u_{m,x} &= L_m \frac{di_x}{dt} + R_m i_x + \frac{2K_m}{r_m}\dot{x} \\ u_{m,y} &= L_m \frac{di_y}{dt} + R_m i_y + \frac{2K_m}{r_m}\dot{y} \end{aligned}} \tag{1.25}$$

The original motor voltages then follow from (1.20):

$$u_{m,L} = \tfrac{1}{2}\left(u_{m,x} + u_{m,y}\right), \qquad u_{m,R} = \tfrac{1}{2}\left(u_{m,x} - u_{m,y}\right) \tag{1.26}$$

Since (1.25) contains $\frac{di_x}{dt}$, and (1.24) contains $\ddot{x}$, the voltages depend on the third derivative of the position. Therefore, for any C^3 functions $x(t)$ and $y(t)$ describing the position of the tool head, we can derive an open-loop analytical solution for the input voltages, even if those voltages are not realistic.

Ellipse

An ellipse with centre (h, k) and semi-axes a and b has the parametric form:

$$x(t) = h + a\cos t, \qquad y(t) = k + b\sin t \tag{1.27}$$

We also need the derivatives up to the third order:

$$\begin{aligned} \dot{x} &= -a\sin t, \quad \ddot{x} = -a\cos t, \quad \dddot{x} = a\sin t \\ \dot{y} &= b\cos t, \quad \ddot{y} = -b\sin t, \quad \dddot{y} = -b\cos t \end{aligned}$$

Substituting into (1.24):

$$\begin{aligned} i_x(t) &= -\frac{m_{\text{eq}} r_m}{K_m} a\cos t - \frac{b_x r_m}{K_m} a\sin t = -\frac{r_m}{K_m} a\left[m_{\text{eq}}\cos t + b_x \sin t\right] \\ i_y(t) &= \frac{r_m}{K_m} b\left[b_y \cos t - m_{\text{eq}}\sin t\right] \end{aligned}$$

And into (1.25):

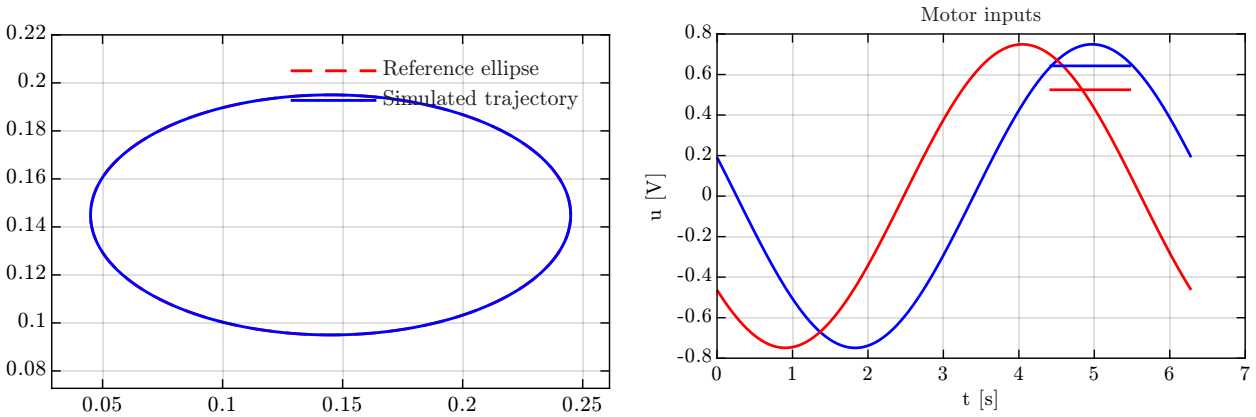
$$\begin{aligned} u_{m,x}(t) &= L_m \frac{r_m}{K_m} a\left[m_{\text{eq}}\sin t - b_x\cos t\right] - R_m \frac{r_m}{K_m} a\left[m_{\text{eq}}\cos t + b_x \sin t\right] - \frac{2K_m}{r_m} a\sin t \\ u_{m,y}(t) &= -L_m \frac{r_m}{K_m} b\left[m_{\text{eq}}\cos t + b_y \sin t\right] + R_m \frac{r_m}{K_m} b\left[b_y\cos t - m_{\text{eq}}\sin t\right] + \frac{2K_m}{r_m} b\cos t \end{aligned}$$

With

$$u_{m,L}(t) = \tfrac{1}{2}\left(u_{m,x}(t) + u_{m,y}(t)\right), \qquad u_{m,R}(t) = \tfrac{1}{2}\left(u_{m,x}(t) - u_{m,y}(t)\right)$$

we obtain the analytical open-loop solution for drawing the ellipse.

Simulating the given system with parameters defined in Table 1, we get near-perfect alignment with the reference ellipse.



(a) Reference and simulated trajectory.

(b) Open-loop motor voltages.

Figure 3: Open-loop drawing of an ellipse with nominal model parameters.

As this is open-loop control, this good result relies on a good parameter estimation. If we change the parameters of the model by ±5% we can see that an error immediately arises.

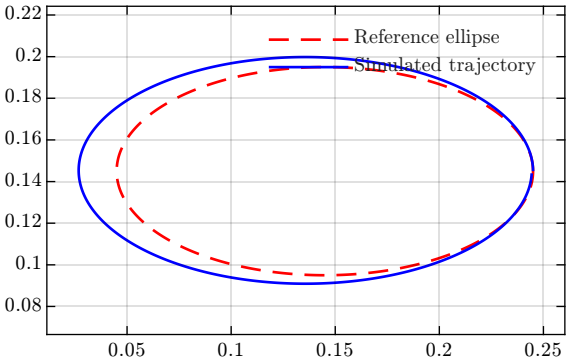
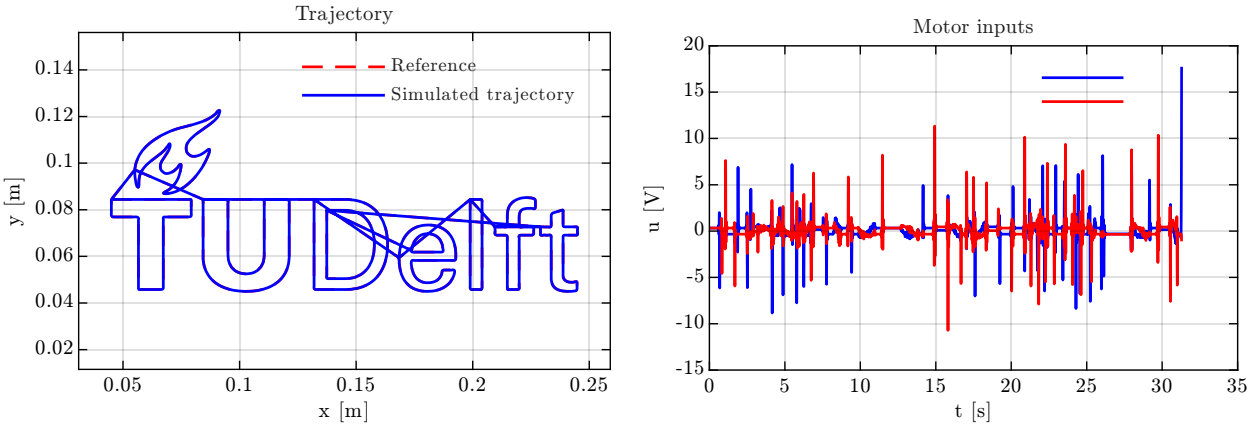


Figure 4: Open-loop drawing of an ellipse with model parameters perturbed by ±5%.

Arbitrary Drawing

As stated in the derivation, any C^3 function used for $x(t)$ and $y(t)$ will allow us to get an analytical solution. With that in mind we can draw arbitrary shapes by approximating them with quartic splines. Using MATLAB's *bwboundaries* function to detect edges of shapes, in combination with *spapi* which gives us an interpolating quartic spline given a set of points, we can draw the *TU Delft* logo[1].

[1]Different letter boundaries are connected by straight lines so the whole drawing can be one input function



(a) Reference and simulated trajectory.

(b) Open-loop motor voltages.

Figure 5: Open-loop drawing of the *TU Delft* logo with nominal model parameters.

Again, like with the ellipse, the logo, because of its sharp corners, is even more sensitive to inaccuracies in the model parameters. If we again perturb the parameters by ±5% we see that control of this form would not be useful anymore.

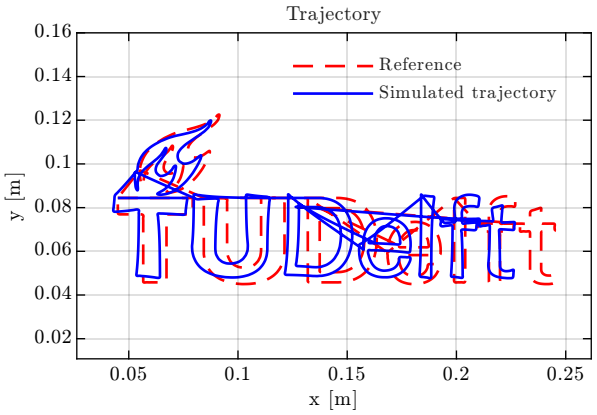


Figure 6: Open-loop drawing of the *TU Delft* logo with model parameters perturbed by ±5%.

1.3.1 Modelling the Nozzle

In the following section we model the nozzle of the 3D printer (Figure 7).

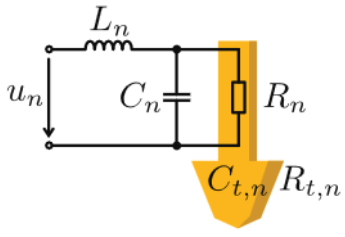


Figure 7: Schematic of the nozzle heating circuit.

The system consists of an electrical and thermodynamical part. A circuit consisting of an

inductor L_n, capacitor C_n and a heating resistor R_n. We choose the current through the inductor and the voltage across the capacitor, $i_{L,n}$ and u_c respectively, as first two states of our system. By solving the three Kirchhoff equations we get the following two first-order differential equations:

$$\begin{aligned} L_n \frac{di_{L,n}}{dt} &= u_n - u_c \\ C_n \frac{du_c}{dt} &= i_{L,n} - \frac{u_c}{R_n(T_r)} \end{aligned} \tag{1.28}$$

Next, we model the thermal part. Besides the thermal capacitance of the nozzle $C_{t,n}$, given in the task, we also model the thermal capacitance of the resistor C_r[2]. We choose two additional states, temperature of the resistor (T_r) and the temperature of the nozzle (T_n). Losses happening between the resistor and the nozzle are described by $R_{t,n}$, and the heat dissipation to the environment and to the filament is modelled with one variable R_{env}. The connection to the electrical part is the heat dissipated by the resistor $P_r = \frac{u_c^2}{R_n(T_r)}$. This gives us the following two equations which with (1.28) fully describe the system:

$$\begin{aligned} C_{t,n} \frac{dT_n}{dt} &= \frac{T_r - T_n}{R_{t,n}} - \frac{T_n - T_e}{R_{\text{env}}} \\ C_r \frac{dT_r}{dt} &= P_r - \frac{T_r - T_n}{R_{t,n}} \end{aligned} \tag{1.29}$$

with

$$R_n(T_r) = R_0 \left[1 + \alpha \left(T_r - T_0\right)\right] \tag{1.30}$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

Due to the square voltage term and the temperature dependence of the resistance the system is very non-linear, although the very different time constants of two systems (thermal states change much slower compared to the electrical) can be exploited. Alternatively, the model can reasonably be simplified by assuming a constant resistance. With that formulation we would get two linear systems connected by a single non-linear term.

[2] If we do not model the thermal capacitance of the resistor, we end up with a differential-algebraic equation (DAE).

1.3.2 Printing a circle of filament

In this section a 1D distributed parameter model for the cooling of a square cross-section filament on a heating plate and a numerical formulation of the problem are derived.

Modelling the system

We take a cross-section of $a \times a$ with length ΔL. Temperature is assumed uniform across the $a \times a$ cross-section to keep the model one dimensional in space. Three faces exchange heat with the environment (T_e), and one face with the heating plate (T_p).

The problem is modelled using the heat equation, Fourier's law gives the flux (1.33). Writing the energy density as (1.32) and substituting both into the energy balance yields the heat equation (1.34):

$$\partial_t u = -\nabla \cdot \phi + f \tag{1.31}$$
$$u = \rho c\, T \tag{1.32}$$
$$\phi = -\kappa \nabla T \tag{1.33}$$

$$\partial_t T = \frac{\kappa}{\rho c}\, \nabla^2 T + \frac{f}{\rho c} \tag{1.34}$$

Secondly, we need to describe the heat exchange of the segment ΔL. Using Newton's cooling law we can express the heat transfer through the surfaces as (1.35). Dividing that by V gives us the heat transfer per unit volume (1.36).

$$\Delta A_e = 3a\, \Delta L, \qquad \Delta A_p = a\, \Delta L, \qquad V = a^2\, \Delta L$$

$$\dot{Q} = K_e\, \Delta A_e\, (T_e - T) + K_p\, \Delta A_p\, (T_p - T) \tag{1.35}$$
$$f = \frac{\dot{Q}}{V} = \frac{1}{a}\Big[\, 3K_e\, (T_e - T) + K_p\, (T_p - T)\,\Big] \tag{1.36}$$

Substituting everything back into (1.34), gives us the final PDE:

$$\boxed{\partial_t T = k\, \partial_{xx} T + \alpha\, T + \beta} \tag{1.37}$$

$$k = \frac{\kappa}{\rho c}, \qquad \alpha = -\frac{3K_e + K_p}{a\, \rho c}, \qquad \beta = \frac{3K_e T_e + K_p T_p}{a\, \rho c}$$

Front Fixing

Due to needing to solve the problem while the nozzle is still moving, this becomes a moving boundary problem. A method from [1] is used. We introduce a transformation ξ, that maps our space to the interval $[0, 1]$, with $s(t)$ being the position of our boundary, and consequently $\dot{s}(t)$ the speed of the boundary/nozzle.

$$\xi = \frac{x}{s(t)}, \qquad 0 \le \xi \le 1, \qquad s(t) = s_0 + s_1 t, \qquad \dot{s} = s_1 \tag{1.38}$$

Applying a change of variable to our equation (1.37) we get:

$$\frac{\partial T}{\partial x} = \frac{1}{s}\frac{\partial T}{\partial \xi}, \qquad \frac{\partial^2 T}{\partial x^2} = \frac{1}{s^2}\frac{\partial^2 T}{\partial \xi^2}$$

$$\left(\frac{\partial T}{\partial t}\right)_x = \left(\frac{\partial T}{\partial t}\right)_\xi + \frac{\partial T}{\partial \xi}\left(\frac{\partial \xi}{\partial t}\right)_x = \left(\frac{\partial T}{\partial t}\right)_\xi - \frac{x\,\dot{s}}{s^2}\frac{\partial T}{\partial \xi} = \left(\frac{\partial T}{\partial t}\right)_\xi - \frac{\xi\,\dot{s}}{s}\frac{\partial T}{\partial \xi}$$

$$\boxed{\left(\frac{\partial T}{\partial t}\right)_\xi = \frac{\xi\,\dot{s}}{s}\frac{\partial T}{\partial \xi} + \frac{k}{s^2}\frac{\partial^2 T}{\partial \xi^2} + \alpha T + \beta} \tag{1.39}$$

Boundary and Initial Conditions

With $T(\xi, t)$ we can place necessary boundary conditions for the problem to be solvable. Firstly, the temperature at the endpoint ($\xi = 1$) at all times equals the temperature of the nozzle which constitutes a Dirichlet condition. Secondly, we assume that the heat dissipation on the ($\xi = 0$) surface is negligible compared to the whole surface, so a Neumann boundary condition is used to model it insulated. Finally, as seen in (1.38), there exists initial length of filament s_0, needed to solve the problem, and its temperature at $t = 0$ is assumed uniform and equal to the nozzle temperature.

$$T(1,t) = T_n, \qquad \partial_\xi T(0,t) = 0, \qquad T(\xi,0) = T_n, \qquad s_0 > 0 \tag{1.40}$$

Discretisation

The temperature is discretised on a uniform grid in the transformed coordinate $\xi \in [0,1]$, with spacing $\Delta\xi = 1/N$, and in time with step Δt. Time is advanced with an implicit (backward) Euler scheme, the diffusion term $\partial_{\xi\xi} T$ uses a second-order central difference, which is symmetric and produces a tridiagonal matrix and the advection term $\partial_\xi T$ uses a first-order upwind difference.

$$\xi_i = i\,\Delta\xi, \quad i = 0, \ldots, N, \qquad t_j = j\,\Delta t, \qquad T_i^j \approx T(\xi_i, t_j)$$

$$\left.\frac{\partial T}{\partial t}\right|_i^j \approx \frac{T_i^j - T_i^{j-1}}{\Delta t}, \qquad \left.\frac{\partial T}{\partial \xi}\right|_i^j \approx \frac{T_{i+1}^j - T_i^j}{\Delta \xi}, \qquad \left.\frac{\partial^2 T}{\partial \xi^2}\right|_i^j \approx \frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta \xi^2} \tag{1.41}$$

$$\frac{T_i^j - T_i^{j-1}}{\Delta t} = \frac{\xi_i\,\dot{s}}{s}\frac{T_{i+1}^j - T_i^j}{\Delta \xi} + \frac{k}{s^2}\frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta \xi^2} + \alpha\, T_i^j + \beta \tag{1.42}$$

With some substitutions and after inserting the approximations into the transformed PDE gives we get:

$$\gamma_i = \frac{\xi_i\,\dot{s}\,\Delta t}{s\,\Delta\xi} = \frac{i\,\dot{s}\,\Delta t}{s}, \qquad g = \frac{k\,\Delta t}{s^2\,\Delta \xi^2}, \qquad s,\ \dot{s} \text{ evaluated at } t_j$$

$$\boxed{T_i^j\left[1 + \gamma_i + 2g - \alpha\,\Delta t\right] = g\,T_{i-1}^j + (\gamma_i + g)\,T_{i+1}^j + T_i^{j-1} + \beta\,\Delta t} \tag{1.43}$$

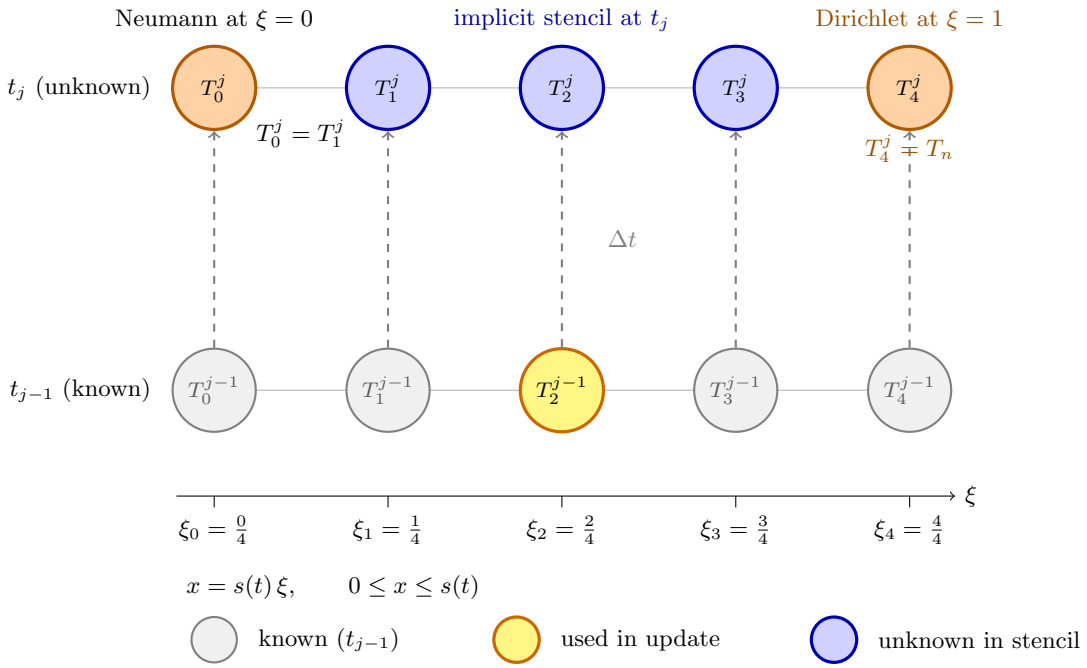


Figure 8: Implicit stencil of (1.43) on the transformed grid ($N = 4$).

We can see that we arrive at an implicit formulation (Figure 8), where we need to solve a linear system at each time step.

Linear System at $t = t_j$

Writing down a couple of terms we can see that we get a tridiagonal matrix, with the first row representing the Neumann condition and the last the Dirichlet condition.

$$d_i = 1 + \gamma_i + 2g - \alpha\,\Delta t, \qquad e_i = \gamma_i + g, \qquad T_0^j = T_1^j, \qquad T_N^j = T_n$$

$$\begin{bmatrix} 1 & -1 & & & & \\ -g & d_1 & -e_1 & & & \\ & -g & d_2 & -e_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -g & d_{N-1} & -e_{N-1} \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ T_2^j \\ \vdots \\ T_{N-1}^j \\ T_N^j \end{bmatrix} = \begin{bmatrix} 0 \\ T_1^{j-1} + \beta\,\Delta t \\ T_2^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-1}^{j-1} + \beta\,\Delta t \\ T_n \end{bmatrix} \tag{1.44}$$

The fact that the matrix is tridiagonal allows us to use specialised solvers like *scipy.linalg.solve_banded*.

After the Boundary Stops (Closed Ring)

After the boundary stops moving, i.e. the circle is closed the matrix needs to be slightly modified, which causes it to lose its tridiagonal form. At the time point of contact the first and last node are connected and become one node, and its temperature is set to the mean of both of them. As the boundary stops moving $\dot{s}$ and consequently γ_i terms disappear.

$$\dot{s} = 0 \;\Rightarrow\; \gamma_i = 0, \qquad s = L, \qquad T_N^j \equiv T_0^j, \qquad g = \frac{k\,\Delta t}{L^2\,\Delta\xi^2}, \qquad d = 1 + 2g - \alpha\,\Delta t$$

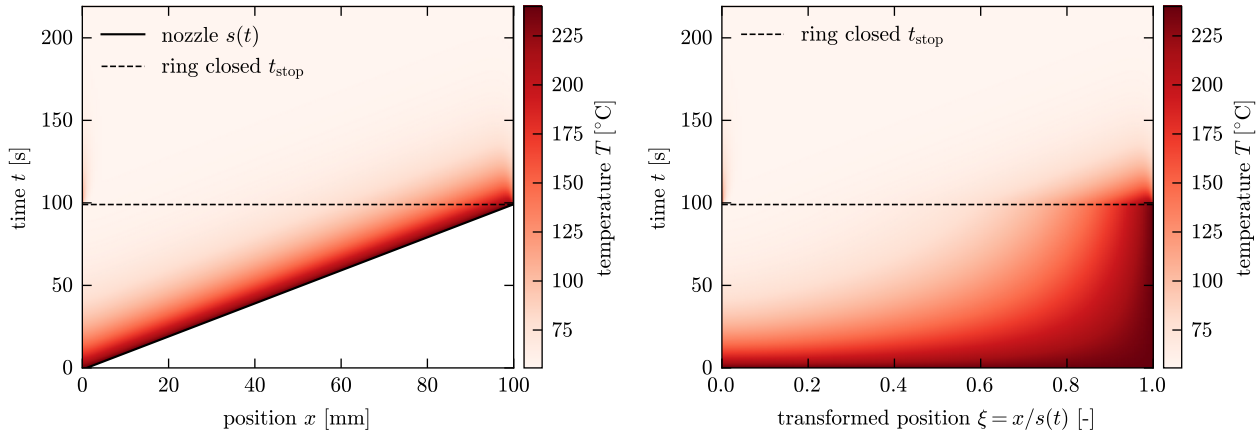
$$\begin{bmatrix} d & -g & & & -g \\ -g & d & -g & & \\ & \ddots & \ddots & \ddots & \\ & & -g & d & -g \\ -g & & & -g & d \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ \vdots \\ T_{N-2}^j \\ T_{N-1}^j \end{bmatrix} = \begin{bmatrix} T_0^{j-1} + \beta\,\Delta t \\ T_1^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-2}^{j-1} + \beta\,\Delta t \\ T_{N-1}^{j-1} + \beta\,\Delta t \end{bmatrix} \tag{1.45}$$

First step after connection:

$$T_0^{j-1} \leftarrow \tfrac{1}{2}\big(T_0^{j-1} + T_N^{j-1}\big)$$

Simulation

The derived model was simulated in *Python* with the parameters listed below.



(a) Physical space x.

(b) Transformed space ξ.

Figure 9: Temperature of the filament while printing and after the ring is closed.

Table 2: Parameters of the filament cooling simulation.

Description	Symbol	Value
Density	ρ	$1240\,\mathrm{kg/m^3}$
Specific heat	c	$1800\,\mathrm{J/(kg\,K)}$
Thermal conductivity	κ	$0.13\,\mathrm{W/(m\,K)}$
Cross-section side	a	$2\,\mathrm{mm}$
Heat transfer coefficient, environment	K_e	$10\,\mathrm{W/(m^2\,K)}$
Heat transfer coefficient, heating plate	K_p	$200\,\mathrm{W/(m^2\,K)}$
Environmental temperature	T_e	$26.85\,^\circ\mathrm{C}$
Heating plate temperature	T_p	$60\,^\circ\mathrm{C}$
Initial filament length	s_0	$1\,\mathrm{mm}$
Nozzle speed	s_1	$1\,\mathrm{mm/s}$
Ring length	L	$100\,\mathrm{mm}$
Nozzle temperature	T_n	$240\,^\circ\mathrm{C}$
Number of intervals	N	200
Time step	Δt	$0.05\,\mathrm{s}$
Simulated time after closing	–	$120\,\mathrm{s}$

2 Finite State Machine

3 Hybrid System

References

[1] John Crank. *Free and Moving Boundary Problems*. Clarendon Press, Oxford, 1984.